

custom_recipes

Custom Recipes

We will load a phylogenetic tree in Newick format for demonstration:

Prepare session:

```
library(ggtree)
library(treeio)
library(ggplot2)
library(dplyr)

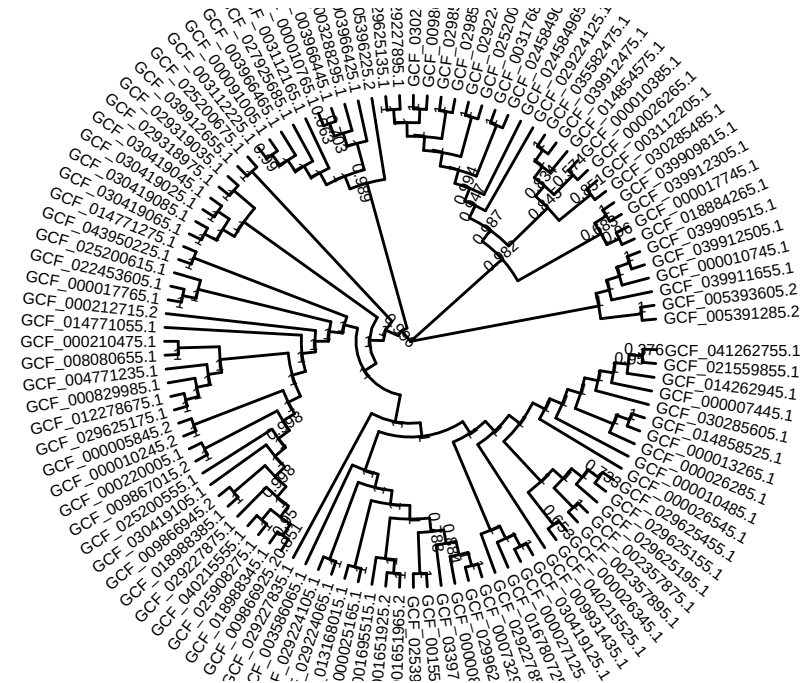
# Load a tree file
tree <- read.newick("example_2_phylo.nwk")
tree$node.label <- round(as.numeric(tree$node.label), digits = 3)
```

Circular Layout Visualization

Circular layouts are excellent for presenting large or complex trees. They allow for efficient space usage and a visually appealing presentation.

```
# Circular tree with basic styling
circular_tree <- ggtree(tree, layout = "circular",
                        branch.length = "none") +
  geom_tiplab(size = 2, aes(angle = angle), offset = 0.5) +
  geom_nodelab(color = "black", size = 2, nudge_y = 0.5)

circular_tree
```

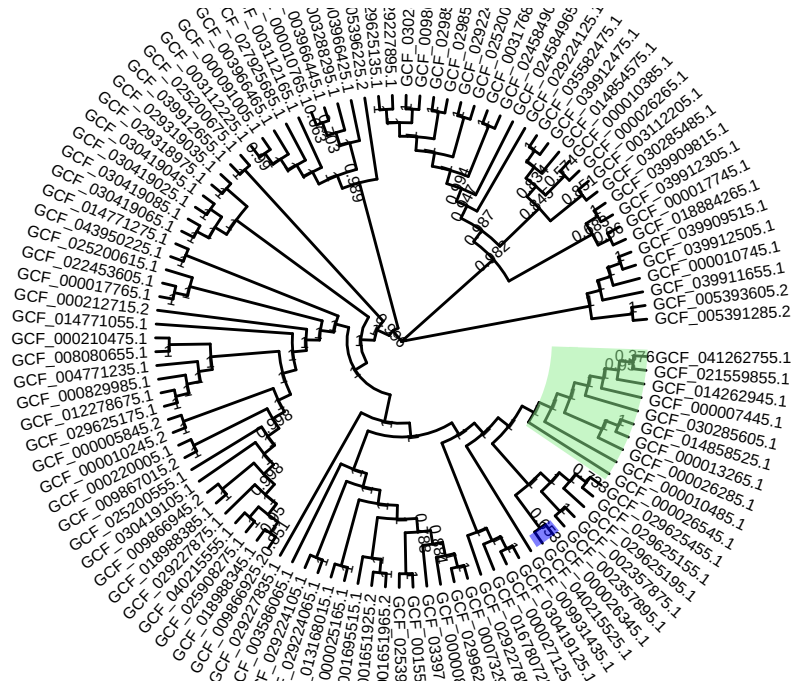


Highlighting Clades in Circular Trees

Highlighting specific clades can make the tree more interpretable. Use the `geom_highlight` function:

```
# Highlight specific clades
highlight_tree <- circular_tree +
  geom_highlight(node = 199, fill = "blue", alpha = 0.5) +
  geom_highlight(node = 200, fill = "lightgreen", alpha = 0.5)

highlight_tree
```



Adding Heatmaps to Circular Trees

We can add metadata as heatmaps around the tree:

```
# Cargar metadata
metadata <- readr::read_tsv("Experiment_Design_example_2.tsv")
metadata <- metadata %>%
  rename(label = Assembly.Accession)

# The rows have to be in the same order as the tree
metadata_ordered <- metadata %>%
  filter(label %in% tree$tip.label) %>%
  arrange(match(label, tree$tip.label))

metadata_ordered$Phylogroup <- replace(metadata_ordered$Phylogroup, is.na(metadata_ordered$Phylogroup), "Unclassified")

metadata_ordered <- metadata_ordered[,c(-1,-2)]
rownames(metadata_ordered) <- make.unique(metadata_ordered$Strain)
```

Warning: Setting row names on a tibble is deprecated.


```
# Applying a theme with a clean background
styled_tree <- p_circular_heatmap +
  theme_tree2() +
  theme(
    plot.background = element_rect(fill = "darkgrey", color = NA),
    panel.border = element_blank(),
    panel.grid = element_blank(),
    axis.text = element_blank(),
    axis.ticks = element_blank()
  )

styled_tree
```



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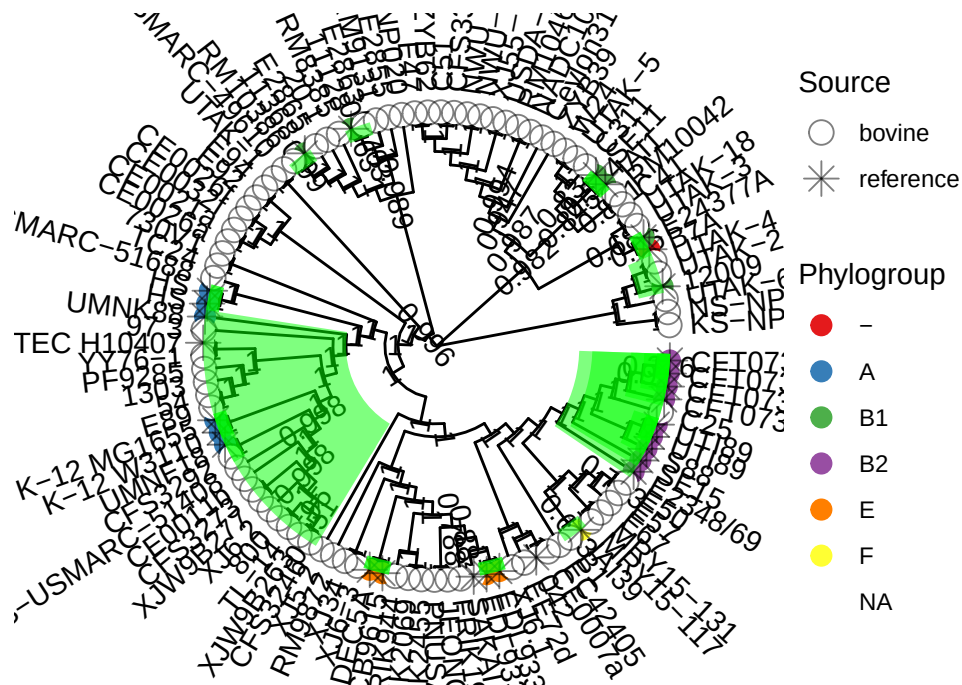
Adding Multiple Aesthetic Layers

Combine multiple layers for richer visuals:

```
# Adding branches with customized colors
pal <- c(
  RColorBrewer::brewer.pal(6, "Set1"),
  "#FFFFFF"
)

p <- ggtree(tree, ladderize = TRUE, branch.length = "none", layout = "circular") %<+% meta
  geom_tiplab(aes(label = label, angle = angle), linesize=.1, size = 4, align = FALSE, col
    offset = 1.5) +
  geom_nodelab(size = 4, color = "black", nudge_x = 0.1) +
  geom_tiptpoint(aes(color = Phylogroup), size = 5, shape = 20) +
  geom_tiptpoint(aes(shape = source), size = 4, alpha = 0.5) +
  scale_shape_manual(values = c(1, 8)) + labs(shape = "Source") +
  scale_color_manual(values = pal) +
  geom_hilight(node = tips_and_nodes$parent, fill = "green", alpha = 0.5)
```

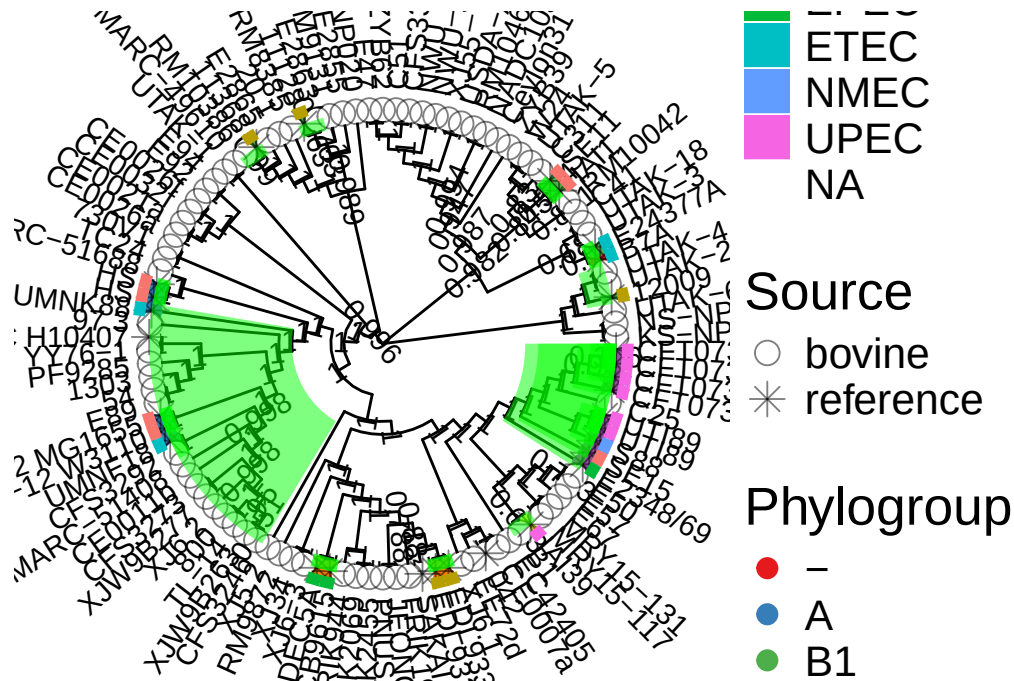
p



Add heatmap

```
matrix_data <- data.frame(Patotype = metadata_ordered$Patotype)
rownames(matrix_data) <- make.unique(metadata_ordered$label)

# Add heatmap
gheatmap(p, matrix_data, width=0.05,
         colnames=FALSE, legend_title="Patotype", color = NA, ) +
  theme(text = element_text(size = 20))
```



Combining Trees with ggnewscale

Use the `ggnewscale` package to add multiple scales of colors:

```
# Adding new scale for different metadata layers
multi_scale_tree <- ggtree(tree, layout = "circular", branch.length = "none") %<+% metadata
  geom_tippoint(aes(color = Patotype), size = 5) +
  scale_color_brewer(palette = "Set2", name = "Patotype") +
  ggnewscale::new_scale_color() +
  geom_tippoint(aes(fill = Phylogroup), shape = 21, size = 3, color = "black") +
```



```

scale_fill_brewer(palette = "Set1", name = "Phylogroup") +
theme(
  legend.position = "right",
  legend.title = element_text(size = 12),
  legend.text = element_text(size = 10),
  plot.title = element_text(size = 14, hjust = 0.5, face = "bold")
) +
ggtitle("Tree with Multiple Aesthetic Layers")

multi_scale_tree

```

Warning: Removed 83 rows containing missing values or values outside the scale range (`geom\_point\_g\_gtree()`).

Tree with Multiple Aesthetic Layers

